

CISCO SECURE WORKLOAD

NIST SP 800-207 Zero Trust Architecture

Zero Trust Tenets & Deployment Model Alignment

Prepared for: [Customer Name]

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Regulatory Standard: NIST SP 800-207 (Zero Trust Architecture — Seven Tenets)

Environment: Hybrid (On-Premises + AWS / Azure / GCP)

Deployment Stage: Fresh Installation

How to read the Coverage column

Direct means Secure Workload produces the primary evidence artifact for that row — segmentation, policy, flow and process telemetry, inventory, vulnerability and reachability, or forensic reconstruction. Direct is still evidence, not an attestation.

Supporting means Secure Workload contributes an input that feeds or complements another control, tool, or process — detection input only, reconciled against an external register, or one log source among several.

Evidence Required means Secure Workload has no record for that row. The customer supplies the contract, the analysis, or the HR / physical-security evidence from another control.

Every report in this set now uses the same three coverage words — Direct, Supporting, and Evidence Required. Coverage describes what evidence Secure Workload produces; it never means the control is passed.

Executive Summary

NIST SP 800-207 defines Zero Trust Architecture (ZTA) as a set of cybersecurity paradigms that move defenses from static, network-based perimeters to focus on users, assets, and resources. The standard defines seven foundational tenets that any ZTA implementation must satisfy, along with three deployment models: Agent-Based, Enclave-Based, and Resource Portal-Based.

Cisco Secure Workload (CSW) is a foundational ZTA technology that directly satisfies NIST 800-207 tenets related to resource protection, communication integrity, dynamic policy enforcement, and continuous monitoring. CSW's workload-level micro-segmentation, Application Dependency Mapping (ADM), vulnerability detection, and forensic telemetry make it the enforcement and visibility layer of a NIST-compliant ZTA implementation.

ZTA Tenet Coverage Summary

NIST 800-207 Tenet	CSW Role	Coverage
Tenet 1 — All data sources and computing services are resources	Workload discovery and inventory across hybrid environments	Direct
Tenet 2 — All communication is secured regardless of network location	Encrypted protocol enforcement, unencrypted flow detection	Direct
Tenet 3 — Access to individual enterprise resources is granted on a per-session basis	Workload-level allowlist policies, session-scoped access enforcement	Direct
Tenet 4 — Access to resources is determined by dynamic policy	ADM-derived policies, API-driven dynamic policy updates	Direct
Tenet 5 — Enterprise monitors and measures integrity and security posture of all assets	Continuous vulnerability scanning, process hash verification, telemetry	Direct
Tenet 6 — All resource authentication and authorization is dynamic and strictly enforced	Encrypted auth path enforcement (LDAPS/Kerberos), unauthenticated access blocking	Supporting
Tenet 7 — Enterprise collects information about current state of assets and uses it to improve security posture	Full flow + process telemetry, ADM baseline, SIEM integration	Direct

NIST 800-207 Tenet-by-Tenet Mapping

Tenet 1 — All Data Sources and Computing Services Are Resources

NIST Requirement: Every device, service, and data source — regardless of ownership or location — must be treated as a resource and managed accordingly.

CSW Implementation	Detail
Workload Discovery	CSW software sensors auto-register every monitored workload as a managed resource. Cloud connectors extend this to AWS EC2, Azure VMs, and GCP instances via agentless discovery.
Universal Inventory	ADM builds a real-time inventory of all computing resources — hostname, OS, processes, open ports, network interfaces, and communication partners.
Label-Based Classification	Every resource is labeled by environment, data classification, application, and owner — establishing the resource identity required by NIST 800-207.
Cloud Resource Parity	Cloud workloads are managed under the same policy framework as on-prem. No implicit trust for cloud-hosted resources — all treated as untrusted until verified.

Evidence generated: Workload inventory export, ADM resource map, cloud connector inventory report.

Tenet 2 — All Communication Is Secured Regardless of Network Location

NIST Requirement: All traffic — including traffic on internal networks — must be protected. Network location alone does not grant implicit trust. Communications must be authenticated and encrypted.

CSW Implementation	Detail
Unencrypted Flow Detection	CSW continuously monitors all workload communications and flags unencrypted protocols: HTTP (port 80), FTP (port 21), Telnet (port 23), plain LDAP (port 389), and legacy TLS versions (1.0/1.1).
Encrypted Path Enforcement	CSW policies enforce HTTPS, SFTP, LDAPS, and SSH as the only permitted protocols for sensitive workloads. Non-compliant connections are blocked and alerted.
Internal Traffic Visibility	ADM captures all east-west (internal) traffic — eliminating the assumption that internal network communication is safe. Every internal flow is validated against the approved policy baseline.
No Implicit Internal Trust	CSW default-deny policies treat internal traffic identically to external traffic. All connections must match an explicit allowlist rule regardless of source network location.

Evidence generated: Protocol compliance report (unencrypted flow detection), policy enforcement log, ADM east-west flow map.

Tenet 3 — Access to Individual Resources Is Granted Per-Session

NIST Requirement: Access must be granted to specific individual resources — not broad network segments — and must be evaluated per connection or session.

CSW Implementation	Detail
Workload-Level Allowlist	CSW policies operate at the individual workload level — not subnet or VLAN. Each workload has its own explicit allowlist. Access to Workload A does not imply access to Workload B, even if they share a subnet.
Per-Flow Policy Evaluation	Every network connection is evaluated against the policy engine in real time. ADM-baseline policies define exactly which source workloads can reach which destination workloads on which ports.
Micro-Segmentation	CSW enforces microsegmentation between every workload pair. Lateral movement — a resource reaching a workload it has no approved flow to — is blocked automatically.
Session Audit Trail	Every connection attempt (approved or denied) is recorded in CSW forensic telemetry with full context: source, destination, process, user, timestamp, and policy decision.

Evidence generated: Per-session flow log, policy denial log, micro-segmentation policy workspace export.

Tenet 4 — Access Is Determined by Dynamic Policy

NIST Requirement: Access policy must be dynamic — incorporating observable state of the requesting resource, user behavioral attributes, and environmental conditions. Static firewall rules are insufficient.

CSW Implementation	Detail
ADM-Derived Policy Baseline	CSW policies are derived from actual observed application behavior (ADM) — not manually authored static rules. The baseline reflects real traffic patterns and is updated as applications evolve.
Label-Driven Dynamic Scoping	As workload labels change (e.g., a workload is classified as PHI-bearing), CSW scope membership — and therefore the policies applied to it — updates automatically without manual rule changes.
API-Driven Policy Updates	CSW REST API allows SOAR platforms, CMDB systems, and orchestration tools to push policy updates in real time. A workload flagged as compromised by EDR triggers an automatic CSW isolation policy via API.
Vulnerability-Informed Policy	CSW vulnerability data drives compensating control policies: a workload with a critical unpatched CVE can be automatically restricted to admin-only access until patched — posture-aware access control.
Anomaly-Triggered Adjustment	CSW anomaly detection identifies behavioral deviations from the ADM baseline. These signals trigger SOAR-mediated policy tightening — moving access decisions from static to continuously adaptive.

Evidence generated: ADM policy workspace, API audit log, anomaly alert-to-policy-change correlation log.

Tenet 5 — Enterprise Monitors and Measures Integrity and Security Posture of All Assets

NIST Requirement: The enterprise must continuously monitor the security posture of all owned and associated assets. No asset is assumed healthy without verification.

CSW Implementation	Detail
Continuous Vulnerability Scanning	CSW continuously scans all monitored workloads for CVEs without requiring a separate scanning tool. Results are CVSS-scored, workload-attributed, and exportable for risk register input.
Process Hash Verification	CSW monitors the hash of all running processes per workload. Deviations from the approved process baseline — new or modified executables — are flagged immediately as integrity violations.
Configuration Drift Detection	CSW detects new listening ports, new processes, and new network connections that deviate from the ADM-established baseline. Any drift from approved configuration triggers an alert.
Real-Time Posture Feed	CSW posture data (vulnerability state, process baseline, network compliance) is continuously streamed to SIEM — providing the asset health data feed that NIST 800-207 requires for dynamic policy decisions.

Evidence generated: Vulnerability scan report (continuous), process anomaly alert log, configuration drift report, SIEM posture feed.

Tenet 6 — Authentication and Authorization Are Dynamic and Strictly Enforced

NIST Requirement: Authentication and authorization must be performed before access is granted and must be re-evaluated continuously. This applies to both human users and non-person entities (NPEs).

CSW Implementation	Detail
Encrypted Auth Path Enforcement	CSW enforces LDAPS (port 636) and Kerberos as the only permitted authentication paths to sensitive workloads. Plain LDAP (port 389) and unauthenticated access attempts are blocked by policy.
Non-Person Entity (NPE) Authentication	CSW captures process-identity context per workload connection — identifying which service or application initiated each flow. Unrecognized process-to-resource connections trigger alerts.
Unauthenticated Access Detection	CSW forensic telemetry identifies connections that bypass the approved authentication path — a direct indicator of credential compromise or lateral movement.
Complementary Controls Required	Full dynamic authentication (MFA re-evaluation, continuous session validation, IdP integration) requires pairing CSW with an Identity Provider (IdP), SSO platform, and/or Cisco Secure Access (ZTNA).

Evidence generated: Authentication path enforcement log (LDAPS policy), unauthenticated access alert log, process-identity flow report.

Note: Tenet 6 is Supporting coverage because CSW enforces the network path to authentication systems but does not itself perform identity provider functions. Pair with Cisco Secure Access or equivalent ZTNA for full coverage.

Tenet 7 — Enterprise Collects Information and Uses It to Improve Security Posture

NIST Requirement: The enterprise must collect, aggregate, and analyze data about the current state of assets, network traffic, and access requests, and use it to improve security posture over time.

CSW Implementation	Detail
Full Flow & Process Telemetry	CSW collects comprehensive telemetry: every network flow (source, destination, port, protocol, bytes, duration), every process (name, hash, parent, user), and every policy decision across all monitored workloads.
ADM Behavioral Baseline	Quarterly ADM re-runs establish an evolving behavioral baseline. Delta analysis between ADM runs reveals new application dependencies, removed flows, and emerging attack surface — enabling proactive posture improvement.
Vulnerability Trend Analysis	CSW tracks CVE emergence and remediation over time per workload. Trending data shows whether the organization's patch posture is improving, stable, or degrading.
SIEM Integration for Analytics	All CSW telemetry streams to SIEM via Kafka/Syslog — enabling the enterprise analytics platform required by NIST 800-207 to correlate CSW data with identity, endpoint, and cloud signals.
Compliance Reporting	CSW compliance reports feed the security governance process. Control

Loop	mapping evidence, vulnerability trends, and policy drift reports provide the data loop required by 800-207's continuous improvement model.
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Evidence generated: SIEM telemetry feed, ADM delta report (quarterly), vulnerability trend report, compliance posture report.

NIST 800-207 Deployment Models — CSW Positioning

NIST 800-207 defines three ZTA deployment models. CSW plays a role in all three:

Deployment Model	NIST 800-207 Description	CSW Role	Status
Agent/Device-Based ZTA	Policy enforcement agent on each device evaluates access requests based on device posture and identity.	CSW software sensor on each workload enforces network policy and collects posture data. Primary CSW deployment model.	Direct
Enclave-Based ZTA	Resources are grouped into enclaves. Access controlled at the gateway. Intra-enclave traffic is implicitly trusted.	CSW replaces implicit intra-enclave trust with workload-level micro-segmentation. ADM validates all intra-enclave flows — no assumed trust within zones.	Direct
Resource Portal-Based ZTA	A resource portal mediates all access. Users connect to the portal which proxies requests to backend resources.	CSW protects backend resources behind the portal. Enforces that only the portal can reach backend workloads. Detects any bypass of the portal path.	Direct

Evidence Collection for NIST 800-207 Compliance

Evidence Artifact	CSW Source	NIST 800-207 Tenet	Frequency
Workload inventory & resource map	Inventory → Export + ADM	Tenet 1	Monthly
Protocol compliance report (unencrypted flows)	Flow Search (protocol filter)	Tenet 2	Quarterly
Per-session flow & policy decision log	Investigate → Flow Search	Tenet 3	Continuous
ADM policy workspace + change log	Defend → Workspace History	Tenet 4	Per change
Vulnerability scan report (trend)	Investigate → Vulnerability	Tenet 5	Weekly
Process baseline + anomaly alert log	Alerts → Triggered Events	Tenet 5	Real-time
Authentication path enforcement log	Policy Analysis (LDAPS rules)	Tenet 6	Continuous
SIEM telemetry feed + ADM delta report	Platform → Data Export + ADM	Tenet 7	Continuous / Quarterly

Recommended Deployment Roadmap

#	Phase	Action	Timeline
1	Resource Enumeration	Deploy CSW sensors on all workloads. Connect cloud accounts. Build complete resource inventory with labels. (Tenet 1)	Days 1–10
2	Communication Visibility	Run ADM for 2–4 weeks. Map all east-west flows. Identify unencrypted communications. (Tenet 2, 7)	Days 11–35
3	Per-Resource Policy	Build workload-level allowlist policies from ADM. Deploy default-deny for all sensitive scopes. (Tenet 3)	Days 36–50
4	Dynamic Policy Enablement	Integrate CSW API with SOAR. Enable label-driven dynamic scoping. Configure vulnerability-informed compensating controls. (Tenet 4)	Days 51–65
5	Continuous Posture Monitoring	Enable SIEM integration. Configure anomaly detection alerts. Establish weekly vulnerability reporting cadence. (Tenet 5, 7)	Days 66–75
6	Auth Path Enforcement	Enforce LDAPS-only paths. Block plain LDAP, Telnet, FTP to sensitive scopes. Pair with IdP/ZTNA for full Tenet 6 coverage. (Tenet 6)	Days 76–90